higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## MATHEMATICS
## (Second Paper)
### NQF LEVEL 4

(10501064)

**25 February 2019 (Y-Paper)**
**13:00–16:00**

**A nonprogrammable calculator may be used.**

**This question paper consists of 7 pages, 1 answer sheet and a formula sheet of 2 pages.**

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**TIME: 3 HOURS**
**MARKS: 100**

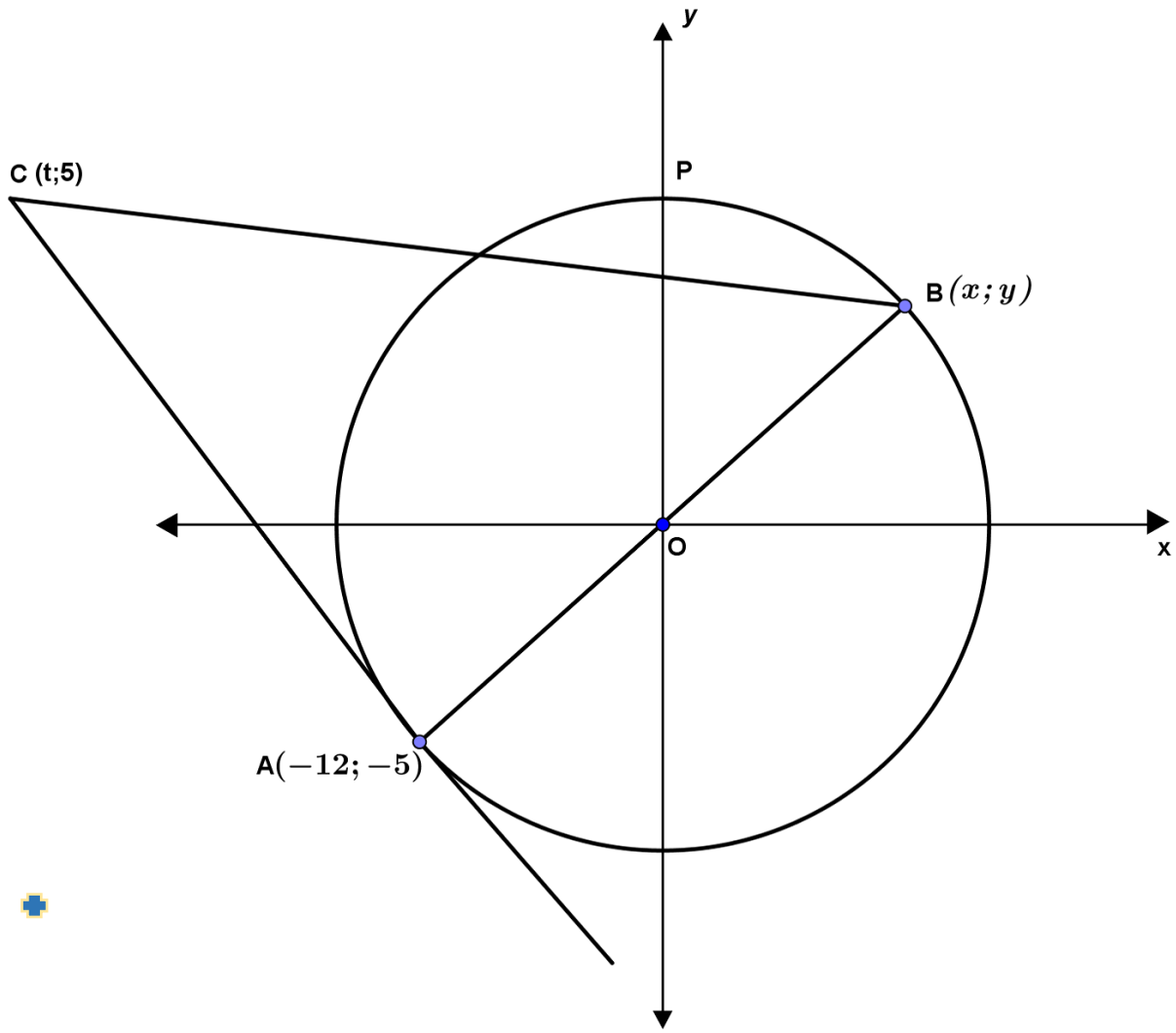
## INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Clearly show ALL the calculations, diagrams, graphs, etc., which you have used in determining the answers.
5. If necessary, answers should be rounded off to THREE decimal places, unless stated otherwise.
6. Diagrams are NOT drawn to scale.
7. Work neatly.

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**QUESTION 1**

1.1 In the figure below, O is the centre of the circle, and A(-12;-5) and B($x$; $y$) are the end points of a diameter of the circle. C is the point ($t$; 5), and CA is a tangent to the circle.




Use the above information to answer the following:

1.1.1 Calculate the length of OA. (1)

1.1.2 Write down the equation of the circle. (1)

1.1.3 Determine the gradient of OA. (1)

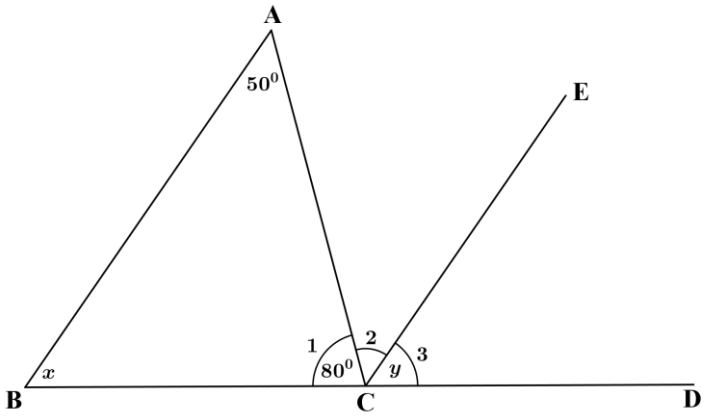
1.1.4 Determine the equation of AC. (3)

1.1.5 Calculate the value of $t$. (2)

1.1.6 Determine the coordinates of B. (2)

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1.2 In the diagram below, side BC of Δ ABC is produced to D. CE bisects $A\hat{C}D$. Also, $\hat{A} = 50^0$, $\hat{C}_1 = 80^0$. Let $\hat{B} = x$ and $\hat{C}_3 = y$.

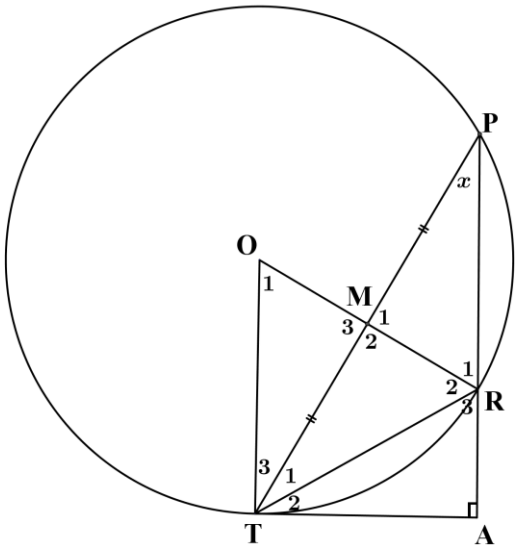



1.2.1 Calculate *x*. (1)

1.2.2 Calculate *y*. (2)

1.2.3 Is CE // BA? Give a reason for your answer. (2)

1.3 In the diagram below, TA is a tangent to circle O at T with points P, R and T on the circle. M is the midpoint of the chord PT. PR is produced to meet TA at A so that TA is perpendicular to PA and OMR is a straight line. OT is drawn. Let $\hat{P} = x$.




1.3.1 Prove that MTAR is a cyclic quadrilateral (2)

1.3.2 Prove that PR = RT (3)

1.3.3 Prove that TR bisects $P\hat{T}A$ (2)

1.3.4 Express $\hat{O}_1$ in terms of *x* (1)

1.3.5 Determine the value of *x* (2)

**[25]**

## QUESTION 2

2.1 Given: cos (A + B) = cos A cos B – sin A sin B.

Prove that:

2.1.1 $\cos 2A = 2\cos^2 A - 1$ (3)

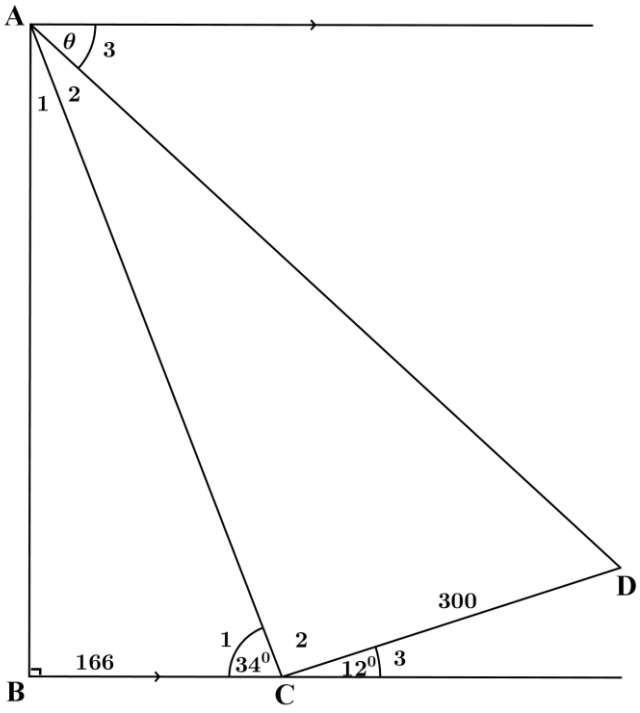
2.1.2 $\cos 2A = 1 - 2\sin^2 A$ (2)

2.2 Determine the values of $x$ in the interval $[0;360^0]$ if $\cos 2x - 7\cos x - 3 = 0$. (5)

2.3 Without using a calculator, determine the value of the following:
$(\sin 75° - \cos 75°)(\sin 75° + \cos 75°)$ (3)

2.4 Determine all values of $\theta$ in the interval $[0^0; 180^0]$ for which $\sqrt{3}\tan 2\theta = 1$. (3)

2.5 A quantity surveyor, at point C, looks up at an angle of 34° to the top of a building AB, 166 m away from him. He turns around to walk in the opposite direction (away from AB) at an inclination of $12^0$, for 300 m to point D. B, C, D and A are on the same vertical plane.




2.5.1 Calculate the size of $\hat{C}_2$. (1)

2.5.2 Calculate the distance AC to the nearest metre. (2)

2.5.3 Calculate the length of AD to the nearest metre. (3)

2.5.4 Hence, or otherwise, determine the size of angle $C\hat{A}D$. (3)

**[25]**

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## QUESTION 3

At a TVET college, 10 learners take Mathematics and Physical Science as subjects. The marks, as a percentage for each learner, are shown in the table below.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Mathematics | 52 | 82 | 94 | 95 | 65 | 77 | 42 | 89 | 48 | 57 |
| Physical Science | 60 | 62 | 88 | 90 | 67 | 75 | 48 | 83 | 57 | 52 |

3.1 On the ANSWER SHEET (attached) make a scatter plot of the marks in the above table. (4)

3.2 Draw a line of best fit for the above data on your graph. (1)

3.3 Calculate the standard deviation of the mathematics data showing all the steps.
NOTE: No marks will be awarded for an answer only. (3)

3.4 Calculate the equation of the least square regression line for the data. (5)

3.5 A learner from this group scored 60% in Mathematics but was absent for the Physical Science examination. Use your equation in question 3.4 to estimate the percentage that this candidate would have scored in Physical Science. (2)

**[15]**

## QUESTION 4

4.1 Define what is meant by *independent events* as related to the concept of probability. (2)

4.2 The following table shows some of the data for a random sample of 100 holidaymakers relating to their preferred destinations by gender, during the December holiday season.

| **Gender** | **Drakensberg National Park** | **Cape Town Table Mountain** | **Pilanesberg Nature Reserve** | **Total** |
|---|---|---|---|---|
| Female | 18 | 16 | --- | 45 |
| Male | --- | --- | 14 | 55 |
| Total | --- | 41 | --- | --- |

4.2.1 Copy and complete the table in your ANSWER BOOK. (3)

4.2.2 Are the events 'being female' and 'preferring the Drakensberg National Park' independent of each other?
Let F represent Female and let D represent Drakensberg National Park. (3)

4.3 A postgraduate student has 3 different graduation photographs labelled A, B and C. However, on her room wall she only has space to hang 2 photographs.

4.3.1 With the aid of a tree diagram, show all the possible ways in which she can arrange 2 photographs. (5)

4.3.2 Calculate the probability that A and C (in any order) will be hanging on the wall. (2)

**[15]**

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## QUESTION 5

5.1 Andi and Zani inherit R50 000 each from their late uncle's estate. Andi will need her money in 4 years' time to fund her studies, while Zani will require her money in 15 years' time.
Bank A offers an interest rate of 8% per annum compounded annually, while bank B offers 12% per annum simple interest.

5.1.1 Show, by calculation, which bank will be the more profitable option for Andi. (5)

5.1.2 Show, by calculation, which bank will be the more profitable option for Zani. (3)

5.2 James completed his 2018 tax return without the assistance of a tax consultant. The 2018 tax year runs from 01 March 2017 to 28 February 2018. He was under 65 years old. He was the main member of a medical aid scheme and he had 3 dependants.

James' total income for the 2018 tax year was R265 000, and his PAYE contributions to SARS (South African Revenue Service) was R2900 per month.

The following amounts were deducted from James' salary **each month**:
R1300 towards his Pension fund (not taxable)
R148,72 towards the Unemployment Insurance Fund (UIF) (not taxable)

Given below are the individual rate tables obtained from (SARS) for the 2018 tax year.

Use the tax tables to calculate how much money is **due to James** or **due to SARS**.

**RATES OF TAX FOR INDIVIDUALS**

| **2018 tax year (1 March 2017 - 28 February 2018)** | |
|---|---|
| **TAXABLE INCOME (R)** | **TAXABLE RATES** |
| 0 – 189 880 | 18% of each R1 |
| 189 881 – 296 540 | 34 178 + 26% of the amount above 189 880 |
| 296 541 – 410 460 | 61 910 + 31% of the amount above 296 540 |
| 410 461 – 555 600 | 97 225 + 36% of the amount above 410 460 |
| 555 601 – 708 310 | 149 475 + 39% of the amount above 555 600 |

**Tax Rebates**

| | **2018** |
|---|---|
| Primary | R13 635 |
| Secondary (65 and older) | R7 479 |
| Tertiary (75 and older) | R2 493 |

**Medical Tax Credit Rates from 2013 tax year onwards**

| **Per month (R)** | **2018** |
|---|---|
| For the taxpayer who paid the medical scheme contributions | R303 |
| For the first dependant | R303 |
| For each additional dependant(s) | R204 |

(12)
**[20]**

**TOTAL: 100**

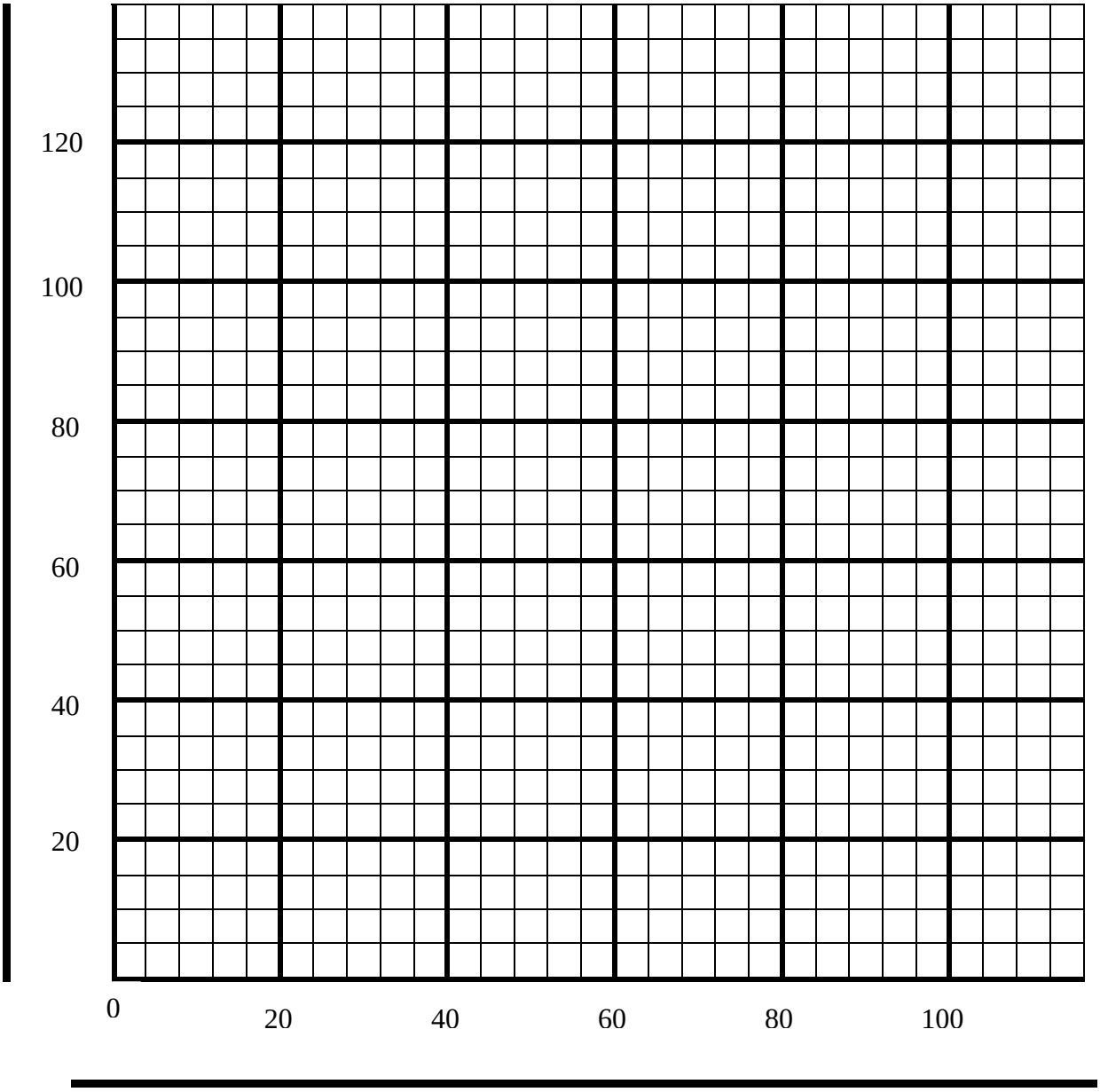
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**ANSWER SHEET**

| **CENTRE NUMBER** | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|

| **EXAMINATION NUMBER** | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

**QUESTION 3.1 and 3.2**




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**FORMULA SHEET**

NOTE: Some of the formulae in this formulae sheet may not be generally true, but may hold under certain conditions. The candidate is required to select the appropriate formulae for a given situation.

1. $y = mx + c$
2. $m = \dfrac{y_2 - y_1}{x_2 - x_1}$
3. $y - y_1 = m(x - x_1)$
4. $(x-a)^2 + (y-b)^2 = r^2$
5. $M\left(\dfrac{x_1 + x_2}{2}; \dfrac{y_1 + y_2}{2}\right)$
6. $m = \tan\theta$
7. $\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
8. $\dfrac{\sin\theta}{\cos\theta} = \tan\theta$
9. $\sin^2\theta + \cos^2\theta = 1$
10. $\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \sin\beta\cos\alpha$
11. $\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$
12. $\sin 2\theta = 2\sin\theta\cos\theta$
13. $\cos 2\theta = \cos^2\theta - \sin^2\theta$
14. $\cos 2\theta = 1 - 2\sin^2\theta$
15. $\cos 2\theta = 2\cos^2\theta - 1$
16. $\dfrac{\sin\hat{A}}{a} = \dfrac{\sin\hat{B}}{b} = \dfrac{\sin\hat{C}}{c}$
17. $a^2 = b^2 + c^2 - 2bc\cos\hat{A}$
18. $\bar{x} = \dfrac{\sum_{i=1}^{n} x_i}{n}$
19. $\text{Variance} = s^2 = \dfrac{\sum(x_i - \bar{x})^2}{n}$

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20. Standard deviation $= \sqrt{\dfrac{\sum (x_i - \bar{x})^2}{n}}$

21. $\hat{y} = a + bx$

22. $b = \dfrac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$ or $b = \dfrac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$

23. $a = \bar{y} - b\bar{x}$

24. $I = A_0 \times \dfrac{r}{100} \times t$ or $I = \dfrac{P\,r\,t}{100}$ or $A = P(1 + in)$

25. $A_t = A_0\left(1 + \dfrac{r}{100 \times m}\right)^{t \times m}$ or $A_t = P\left(1 + \dfrac{r}{100}\right)^n$ or $A = P(1+i)^n$

26. $A_t = A_o\left(1 - \dfrac{r}{100}\right)^t$ or $A = P(1 - in)$ or $A = P(1-i)^n$

27. $i = \dfrac{r}{100}$

28. $\mathrm{P(A)} = \dfrac{n(\mathrm{A})}{n(\mathrm{S})}$

29. $\mathrm{P(A\ and\ B)} = \mathrm{P(A)} \times \mathrm{P(B)}$

30. $\mathrm{P(A\ or\ B)} = \mathrm{P(A)} + \mathrm{P(B)}$

31. $\mathrm{P(A\ or\ B)} = \mathrm{P(A)} + \mathrm{P(B)} - \mathrm{P(A\ and\ B)}$

32. $\mathrm{P(A/B)} = \dfrac{\mathrm{P(A\ and\ B)}}{\mathrm{P(B)}}$

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